

JATIN KUMAR

VIT Vellore
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PROFILE SUMMARY

B.Tech CSE graduate (VIT Vellore) and published researcher (ICCS'25, Singapore) with deep hands-on experience building and evaluating AI agents – Claude, Codex-style tools, LangChain, RAG pipelines. Understands how agents consume context, miss constraints, hallucinate, and drift when prompts are underspecified; has identified and fixed these failure patterns across research (IIT-BHU) and production systems. Communicates technical quality judgments precisely, owns decisions end-to-end, and explains reasoning that others can audit.

EDUCATION

Vellore Institute of Technology (VIT), Vellore, India Sep 2022 – Jul 2026
B.Tech. in Computer Science and Engineering CGPA: 8.64
Relevant Courses: Machine Learning, Artificial Intelligence, Algorithms, Statistics, Computer Vision

Global Indian International School (GIIS), NOIDA, India Apr 2021 – Mar 2022
Higher Secondary Education – 12th Standard Percentage: 92.2

EXPERIENCE

ML Research Intern Sep 2024 – Apr 2025
IIT-BHU, Varanasi

- Evaluated 10+ state-of-the-art continual learning methods across 3 molecular datasets; identified systematic failure patterns (catastrophic forgetting, instability) and designed MTL-FECAM to address them.
- Wrote and refined task specifications for model experiments, ensuring constraints were unambiguous and evaluation criteria were robust – a direct analog to RL task/dataset quality work.
- Achieved **90.89%–92.62%** accuracy, **0.0–0.27** forgetting; published at ICCS'25 (Singapore). [\[Paper\]](#)

Web Developer (Freelance) Jan 2024 – Apr 2024
SAYTRIP WAYS PVT LTD, Remote

- Developed **Tripster**, a live flight booking platform in Next.js; wrote clear component contracts and API specifications with well-defined constraints and expected behaviors.

PROJECTS

Exa.ai Search Memory Agent
Python, LangChain, Exa.ai API, OpenAI API, ChromaDB

- Built an agentic search system with a continual-learning feedback loop; observed agent drift when retrieval context was noisy, identified constraint-following failures in tool-selection logic, and fixed them to improve output consistency across sessions.

Kitaab – Semantic Book Recommender
Python, LangChain, ChromaDB, OpenAI Embeddings, Hugging Face, Gradio

- Reviewed and cleaned 5,200+ book descriptions for accuracy, completeness, and embedding quality; identified data issues degrading retrieval precision and corrected them.
- Evaluated zero-shot classification and DistilRoBERTa emotion tags for correctness; caught systematic misclassifications and revised the pipeline. Live: [Kitaab](#)

Aarogyam – AI Health Diagnostic Tool
Python, PyTorch, Gemini API, Streamlit

- Integrated Google Gemini API for symptom reasoning; evaluated outputs for hallucinations, missing constraints, and contradictory responses across test cases.
- Designed evaluation criteria for diagnostic accuracy; flagged and revised ambiguous prompt templates causing unreliable model behavior. Live: [Aarogyam](#)

SHL Conversational Assessment Recommender Agent
Python, FastAPI, Vector Retrieval, JSON Schema

- Built a conversational AI agent for SHL take-home assessment; designed structured JSON schemas and evaluated agent responses for constraint adherence and output consistency.
- Stress-tested edge cases where the agent missed implicit requirements or overfitted to surface-level patterns.

TECHNICAL SKILLS

AI Agents & LLMs: Claude (Claude Code), OpenAI API (Codex/GPT), LangChain, Agentic AI, RAG, Prompt Engineering, LLM Evaluation, Context Management, Hugging Face Transformers
Languages & Tools: Python, Bash, Git, GitHub, SQL, Linux, Docker (familiar), Jupyter
ML & Data: PyTorch, Numpy, Pandas, ChromaDB (Vector DBs), Statistical Modeling, NLP, Streamlit
Web / APIs: FastAPI, REST APIs, React.js, Next.js, TailwindCSS

ACHIEVEMENTS & RECOGNITION

- **Published Researcher** – ICCS'25, Singapore (Springer LNCS); peer-reviewed paper on continual learning for molecular property prediction
- **Winner** – **Best Fresher Award**, Reverse Coding 2023 by ACM-VIT
- **Top 10 Finalist**, Yuvamanthan Hackathon 2024, organized by IIT Madras